

ECE 310 Spring 2025

Lecture 1: Introduction to Digital Signals

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Welcome to ECE 310!

Let's take a look at the course website.





- Is ECE 310 harder than ECE 210?



ECE 310 Frequently Asked Questions

- Is ECE 310 harder than ECE 210?
- I had a really tough time with ECE 210. Should I be concerned taking ECE 310?



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- Okay, but I also didn't super duper understand Fourier transforms or convolution or etc...

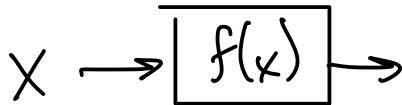


- Is ECE 310 harder than ECE 210?
- I had a really tough time with ECE 210. Should I be concerned taking ECE 310?
- Okay, but I also didn't super duper understand Fourier transforms or convolution or etc...
- What questions do you have?



What is signal processing?

- * Modify data for a desired purpose
- * Any means for representing or expressing data
- * Extracting information
- * Defining behaviour of a system



Learning Objectives

- Describe what defines a signal.
- Explain the difference between continuous-time and discrete-time signals.
- Understand the advantages of working with digital signals over analog signals.



What is a signal?



What is a signal?

- A signal is any measurable quantity that varies over time, space, or any other variable that orders or organizes these measurements.



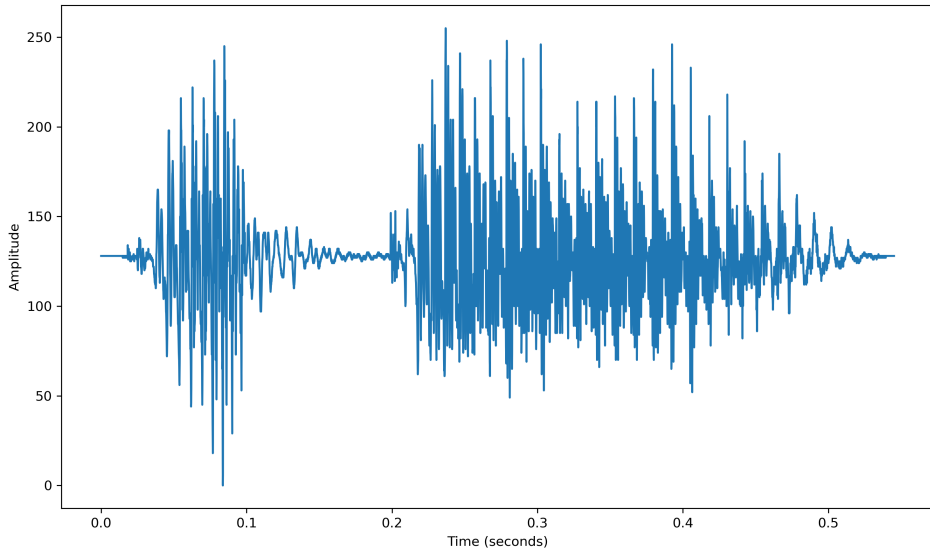
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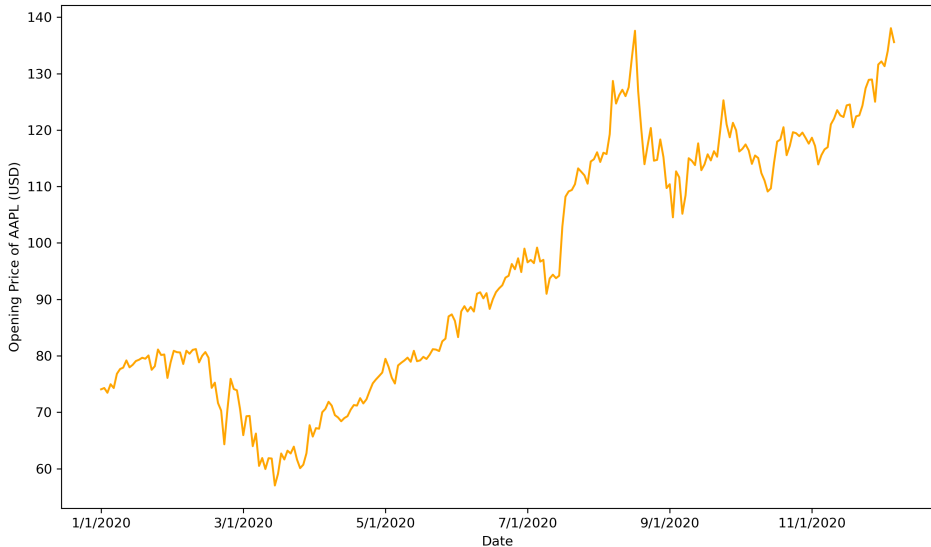
This sounds quite broad, but that's because signal processing is so widely applicable!



Signal Examples: Audio Signal



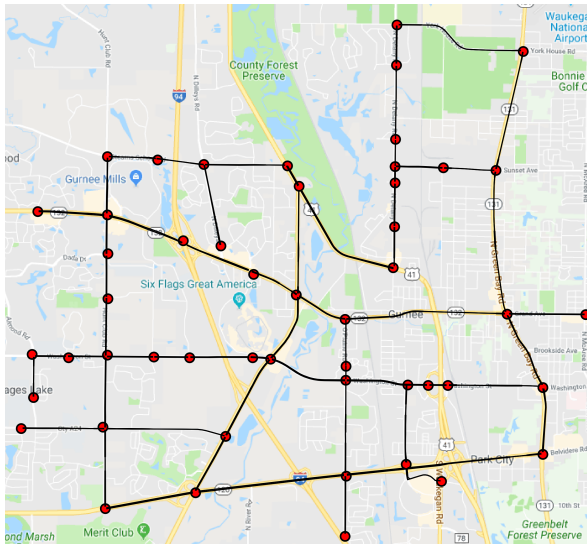
Signal Examples: Financial Signal



Signal Examples: Image Signal



Signal Examples: Graph Signal



Continuous-time vs. Discrete-time Signals



Continuous-time vs. Discrete-time Signals

- A *continuous-time* signal $x(t)$ is observed continuously over time. Every time interval contains infinitely many values from the signal.



Continuous-time vs. Discrete-time Signals

$t \in \mathbb{R} \rightarrow \text{real numbers}$

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- A *discrete-time* signal $x[n]$ is observed at regularly spaced points in time. We say that the time-axis is *discretized*. $n \in \mathbb{Z} \rightarrow \text{integers}$



Continuous-time vs. Discrete-time Signals

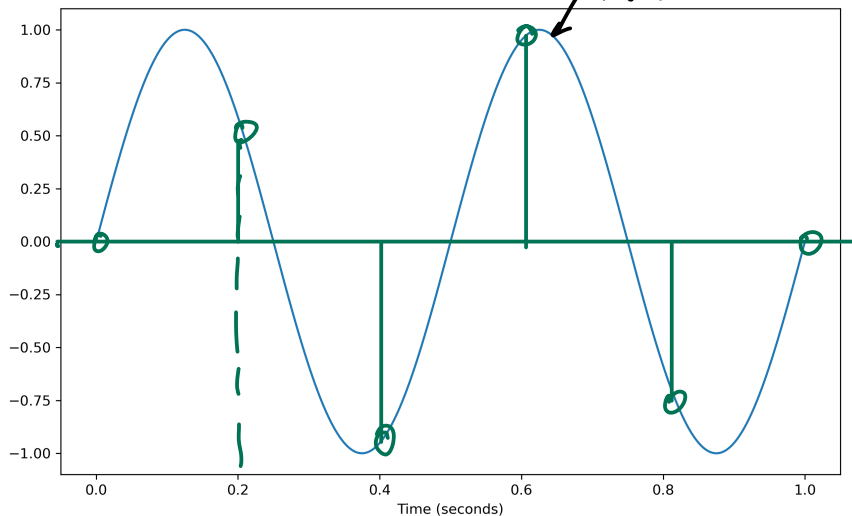
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- For example, audio is commonly recorded at 44.1 kHz. Thus, we have a new sample every $T = 1/44,100$ seconds. T is called the *sampling period*.

$$x[n] = x(nT)$$



Continuous-time vs. Discrete-time Signals

$$T = 0.2s \quad x[n] = x(nT) = x\left(\frac{n}{5}\right)$$



Continuous-valued vs. Discrete-valued Signals



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- A discrete-time, discrete-valued signal is referred to as a *digital signal*.*
 - We will assume our signals are continuous-valued in this class since modern computing systems have very high precision.



Why digital signals?

What are some advantages of using digital signals over analog signals?

1. Digital signals are more reliable.
2. Math is easy. Ex: $e^{x[n]}$
3. Design flexibility
4. Easy to store

